

Final Business Insights Report

E-Commerce Data Analytics — Internship Program

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Dataset: Global E-Commerce | Records: 51,271 | Period: 2011-2014 | Tools: Excel . SQL . Power BI

TOTAL ORDERS 25,029 Unique order IDs	TOTAL SALES \$12,635,222 Across all markets	TOTAL PROFIT \$1,467,498 11.61% margin	LOSS ORDERS 12,538 24.4% of total
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01 EXECUTIVE SUMMARY

This report presents a comprehensive analysis of a global e-commerce dataset containing 51,271 order records across multiple markets, product categories, and customer segments. The analysis was performed in three levels - Excel (Data Cleaning & Analysis), SQL (Querying & Insights), and Power BI (Interactive Dashboard) - as part of the Primeor Solutions Data Analytics Internship Program.

TOTAL REVENUE \$12,635,222 Global sales	TOTAL PROFIT \$1,467,498 Net earnings	PROFIT MARGIN 11.61% Can be improved	LOSS-MAKING ORDERS 12,538 24.4% of all orders
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Market-wise Revenue Breakdown

#	Market	Revenue (\$)	Share %	Status
1	APAC	3,581,785	28.3%	Leader
2	EU	2,937,166	23.2%	Strong
3	US	2,297,022	18.2%	Strong
4	LATAM	2,163,919	17.1%	Growing
5	EMEA	805,053	6.4%	Growing
6	Africa	783,345	6.2%	Emerging
7	Canada	66,932	0.5%	Low

Segment-wise Sales

Segment	Total Sales (\$)	% of Total	Key Insight
Consumer	6,503,476	51.5%	Primary customer base — B2C focus needed
Corporate	3,822,954	30.3%	Stable and consistent revenue stream
Home Office	2,308,792	18.3%	Growth potential in remote work era

02 LEVEL 1 — DATA CLEANING & EXCEL ANALYSIS

Task 1 — Data Cleaning Summary

Cleaning Operation	Action Performed	Status
Duplicate Records	Removed all duplicate order rows from dataset	✓
Missing / Null Values	Filled with 'Unknown' or removed where necessary	✓
Inconsistent Text Values	Standardized Category, Segment, Ship Mode fields	✓
Date Format Issues	Fixed order_date and ship_date to uniform format	✓
Invalid Numerical Values	Corrected negative/zero values in key columns	✓
Extra Spaces & Formatting	Trimmed whitespace from all text columns	✓
Year Column Added	Extracted year from order_date for trend analysis	✓

Task 2 — Basic Excel Analysis Results

Metric	Value
Total Sales	\$12,635,222
Total Profit	\$1,467,498
Average Discount	14.29%
Total Orders	25,029
Total Records	51,271
Profit Margin	11.61%

Top 5 Products by Sales

#	Product Name	Sales (\$)
1	Apple Smart Phone, Full Size	86,936
2	Cisco Smart Phone, Full Size	76,441
3	Motorola Smart Phone, Full Size	73,159
4	Nokia Smart Phone, Full Size	71,904
5	Canon imageCLASS 2200 Advanced Copier	61,600

Top 5 Loss-Making Products

#	Product Name	Total Profit (\$)	Category
1	Cubify CubeX 3D Printer Double Head Print	-8,880	Technology
2	Lexmark MX611dhe Monochrome Laser Printer	-4,590	Technology
3	Motorola Smart Phone, Cordless	-4,447	Technology
4	Cubify CubeX 3D Printer Triple Head Print	-3,840	Technology
5	Bevis Round Table, Adjustable Height	-3,650	Furniture

Task 3 — Pivot Table: Region-wise & Category Sales

Region	Sales (\$)	Category	Profit (\$)	Avg Discount
Central	2,821,740	Technology	663,200	13.54%
South	1,600,600	Office Supplies	518,173	13.74%
North	1,248,121	Furniture	286,125	16.80%
Oceania	1,098,253			
Southeast Asia	883,702			

Monthly Sales Trend

Month	Sales (\$)	Month	Sales (\$)
January	673,535	July	749,423
February	543,768	August	1,293,852
March	769,729	September	1,432,087
April	698,661	October	1,168,220
May	904,061	November	1,551,319 PEAK
June	1,269,751	December	1,580,816 PEAK

Note: November & December are peak sales months — strong Q4 holiday season demand.

10 SQL Queries — Results Summary

#	Query	Key Finding
Q1	Top 10 Profitable Products	Canon imageCLASS 2200 Copier — \$25,200 profit
Q2	Top 10 Customers by Sales	Tom Ashbrook tops with \$40,489 in total sales
Q3	Region-wise Total Sales	Central region leads with \$2,821,740
Q4	Category-wise Avg Profit	Technology has highest avg profit per order
Q5	Highest Discount Category	Furniture — 16.8% average discount
Q6	Orders with Negative Profit	12,538 orders (24.4%) show negative profit
Q7	Monthly Sales Trend	December peaks at \$1,580,816 — holiday season
Q8	Market-wise Revenue Analysis	APAC market tops revenue at \$3,581,785
Q9	Top Performing Sub-Categories	Phones lead with \$1,706,642 in total sales
Q10	Ship Mode Usage Analysis	Standard Class used 30,764 times (59.5%)

Task 3 — Business Insights from SQL Analysis

Q1. Which market generates highest revenue?	APAC — \$3,581,785
The APAC (Asia-Pacific) market generates the highest revenue of \$3,581,785, followed by EU (\$2,937,166) and US (\$2,297,022). APAC's dominance is driven by high order volumes from Australia, India, China, and Southeast Asian countries. Strong consumer demand and a large customer base contribute to its top position. Businesses should prioritize APAC for marketing investment and product expansion.	
Q2. Which category is least profitable?	Furniture — only \$286,125 profit
The Furniture category is the least profitable despite generating significant sales. Two major reasons: (1) It receives the highest average discount of 16.8%, and (2) shipping costs for furniture are substantially higher due to the size and weight of items. Sub-categories like Tables and Bookcases frequently generate negative profit margins. Reducing discounts below 10% could significantly improve profitability.	
Q3. Which shipping mode is most commonly used?	Standard Class — 59.5% (30,764 orders)
Standard Class is by far the most popular shipping mode at 59.5% of all orders (30,764). It is preferred for its cost-effectiveness and wide availability across all regions. Same Day delivery accounts for only 5.2% as it is reserved for urgent/high-priority orders. This distribution reflects good cost-optimization already in place.	

Ship Mode Usage Breakdown

Ship Mode	Order Count	% of Total	Usage Pattern
Standard Class	30,764	59.5%	Most cost-effective — widely preferred by customers
Second Class	10,304	20.1%	Moderate speed and cost balance
First Class	7,504	14.6%	Faster delivery for priority orders
Same Day	2,699	5.2%	Urgent/critical orders only

Top 10 Customers by Sales (Q2)

#	Customer Name	Total Sales (\$)
1	Tom Ashbrook	40,489
2	Tamara Chand	37,453
3	Greg Tran	35,552
4	Christopher Conant	35,187
5	Sean Miller	35,131
6	Bart Watters	32,315
7	Natalie Fritzler	31,778
8	Fred Hopkins	30,404
9	Jane Waco	30,288
10	Hunter Lopez	30,246

04 LEVEL 3 — POWER BI DASHBOARD OVERVIEW

An interactive 3-page Power BI dashboard was created using the cleaned dataset. The dashboard enables dynamic exploration of sales, profitability, customer behaviour, and product performance across markets, categories, and time periods.

Dashboard Pages — Content Overview

Page	Title	Visuals Included
Page 1	Executive Overview	Total Sales KPI Total Profit KPI Total Orders KPI Avg Discount KPI
Page 2	Sales Analysis	Sales by Region (Bar) Sales by Market (Donut) Monthly Trend (Line) Category Performance
Page 3	Product & Customer Insights	Top 10 Products (Bar) Top 10 Customers (Table) Profit by Sub-Category Segment Analysis (Pie)

Dashboard Features Implemented

Feature	Details	Status
Slicers / Filters	Year, Category, Market, Segment slicers on all pages	✓
KPI Cards	4 KPI cards on Executive Overview page	✓
Interactive Visuals	All charts are cross-filterable by slicer selections	✓
Proper Formatting	Labels, titles, axes, legends properly formatted	✓
Consistent Color Theme	Blue theme used consistently across all 3 pages	✓

Top Performing Sub-Categories (Power BI Insight)

#	Sub-Category	Total Sales (\$)	Category
1	Phones	1,706,642	Technology
2	Copiers	1,508,370	Technology
3	Chairs	1,500,930	Furniture
4	Bookcases	1,463,615	Furniture
5	Storage	1,126,579	Office Supplies
6	Appliances	1,011,081	Office Supplies
7	Machines	779,071	Technology
8	Tables	756,031	Furniture
9	Accessories	748,892	Technology
10	Binders	461,842	Office Supplies

05 KEY BUSINESS INSIGHTS & RECOMMENDATIONS

Revenue & Market Performance

- APAC is the top revenue market (\$3.58M) — primary focus area for business growth.
- EU at \$2.94M is a strong second — European consumer demand is consistent year-round.
- Overall profit margin is only 11.61% — controlling discounts in Furniture can improve this to ~14-15%.
- Canada market is significantly underperforming at only \$66,932 (0.5%) — needs strategic review.

Product & Category Analysis

- Technology category is the most profitable at \$663,200 — Phones and Copiers should be prioritized.
- Furniture is least profitable — 16.8% discount and high shipping costs are the main reasons.
- Cubify 3D Printers are biggest loss-makers (-\$8,880 & -\$3,840) — reprice or discontinue immediately.
- Canon imageCLASS 2200 Copier is most profitable single product at \$25,200 — promote heavily.

Customer & Segment Insights

- Consumer segment drives 51.5% of total sales — B2C marketing strategies should dominate budgets.
- Top 10 customers contribute significantly — a loyalty program can improve retention.
- Home Office segment (18.3%) has untapped growth potential in the remote-work era.

Shipping & Operations

- 59.5% of orders use Standard Class — good cost-efficient shipping already in place.
- 24.4% of orders (12,538) have negative profit — mainly due to excessive discounts in Furniture.
- Same Day shipping (5.2%) should be restricted to high-value orders only.

Seasonal & Trend Insights

- Q4 (Oct-Dec) is peak season — revenue spikes in November (\$1.55M) and December (\$1.58M).
- Q1 (Jan-Feb) shows lowest sales — targeted marketing campaigns needed in off-peak months.
- Consistent seasonal patterns year-over-year — useful for inventory and sales forecasting.

Top 3 Actionable Recommendations

1	Reduce Furniture category discounts to below 10% Expected Impact: Increase category profit margin by ~30%
2	Focus marketing and inventory on APAC & EU markets Expected Impact: Maximize ROI in highest-revenue zones
3	Discontinue or reprice Cubify 3D Printer product range Expected Impact: Eliminate top 2 loss-making products, saving ~\$12,720/year

06 CONCLUSION & PROJECT COMPLETION

This internship project provided comprehensive hands-on experience across the complete data analytics pipeline - from raw data cleaning in Excel, to structured business querying in SQL, and finally to interactive BI dashboards in Power BI. The analysis of 51,271 e-commerce records across 7 global markets revealed clear patterns and actionable business insights that can directly improve revenue, reduce losses, and optimize operational efficiency.

Final Project Completion Checklist

Task	Level	Submitted File	Status
Data Cleaning in Excel	Level 1	Cleaned_Dataset.xlsx	✓
Basic Excel Analysis (Sales, Profit, Discount)	Level 1	Excel_Analysis_File.xlsx	✓
Pivot Tables & Charts	Level 1	Excel_Analysis_File.xlsx	✓
SQL Data Import	Level 2	Cleaned_Dataset.xlsx	✓
10 SQL Queries Written & Executed	Level 2	SQL_Queries_File.sql	✓
SQL Business Insights (3 questions)	Level 2	SQL_Queries_File.sql	✓
Power BI 3-Page Dashboard	Level 3	Power_BI_Dashboard.pbix	✓
Dashboard Features (Slicers, KPIs, Visuals)	Level 3	Power_BI_Dashboard.pbix	✓
Final Business Insights Report	Final	Business_Insights_Report.pdf	✓

Submitted Files

File Name	Format	Level	Status
Cleaned_Dataset	.xlsx	Level 1	✓ Ready
Excel_Analysis_File	.xlsx	Level 1	✓ Ready
SQL_Queries_File	.sql	Level 2	✓ Ready
Power_BI_Dashboard	.pbix	Level 3	✓ Ready
Final_Business_Insights_Report	.pdf	Final	✓ Ready